Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 09** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** comprises 6 questions of 1 mark each (5 MCQs and 1 Assertion-Reason).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based integrated unit of assessment of 4 marks.
7. Use of calculators is strictly prohibited. Draw neat, labeled geometric diagrams wherever required.

SECTION A: OBJECTIVE QUESTIONS (6 Marks)

1. If $\Delta ABC \sim \Delta DEF$ and AL, DM are their respective medians, then:
(a) $\frac{AB}{DE} = \frac{AL^2}{DM^2}$ (b) $\frac{AB}{DE} = \frac{AL}{DM}$ (c) $\frac{BC}{EF} = \frac{DM}{AL}$ (d) $\frac{AL}{DM} = \frac{AB^2}{DE^2}$

2. In ΔABC, $DE \parallel BC$ intersects AB at D and AC at E. If $AD = 1.5$ cm, $BD = 4.5$ cm, and $BC = 8$ cm, then DE is:
(a) 2 cm (b) 2.5 cm (c) 3 cm (d) 1.8 cm

3. If a line divides two sides of a triangle in the same ratio, then the line is parallel to the third side. This theorem is:
(a) Thales' Theorem (b) Converse of Thales' Theorem (c) Pythagoras Theorem (d) Midpoint Theorem

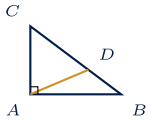
4. Which of the following pairs of triangles are **always** similar?
(a) Two right triangles (b) Two isosceles right triangles (c) Two obtuse triangles (d) Two scalene triangles

5. In ΔABC, $DE \parallel BC$ such that $AD = 2.4$ cm, $AE = 3.2$ cm, and $EC = 4.8$ cm. The length of side AB is:
(a) 6.0 cm (b) 3.6 cm (c) 7.2 cm (d) 5.4 cm

6. **Assertion (A):** In right triangle PQR with $\angle Q = 90^\circ$ and $QS \perp PR$, $QS^2 = PS \cdot SR$.
Reason (R): The altitude to the hypotenuse divides the triangle into two triangles that are similar to each other.
(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (4 Marks)

7. **[2 Marks]** In the given figure, $\angle CAB = 90^\circ$ and $AD \perp BC$. If $AC = 75$ cm, $AB = 100$ cm, and $BC = 125$ cm, find the length of altitude AD.

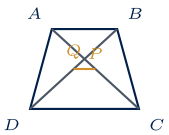


8. **[2 Marks]** In ΔABC and ΔDEF, $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{3}{4}$. If the perimeter of ΔABC is 27 cm, determine the perimeter of ΔDEF.

SECTION C: SHORT ANSWER QUESTIONS (6 Marks)

9. **[3 Marks]** Two trees of heights 7 m and 12 m stand on level ground. If their bases are 12 m apart, find the straight-line distance between their tops.

10. **[3 Marks]** In quadrilateral $ABCD$, a line through D parallel to BC meets AC at P, and a line through C parallel to AD meets BD at Q. Prove that:



$$PQ \parallel AB$$

SECTION D: LONG ANSWER / HOTS QUESTIONS (10 Marks)

11. **[5 Marks]** Prove that if two sides and a median bisecting one of these sides of a triangle are respectively proportional to the two sides and the corresponding median of another triangle, then the two triangles are similar ($\Delta ABC \sim \Delta PQR$ given $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$).

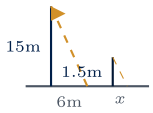
12. **[5 Marks]** Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Prove that:

$$\Delta ABC \sim \Delta PQR$$

SECTION E: COMPETENCY / CASE STUDY QUESTION (4 Marks)

13. **Case Study 09: School Sports Ground Flagpole Calibration & Scale Factor**

During a mathematics lab activity, Class 10 students are tasked with calibrating a scale model for the school sports complex. The school flagpole is 15 m high. At 10:00 AM, its shadow measures 6 m. Nearby, a student of height 1.5 m casts a shadow x m.



(i) **[1 Mark]** Find the exact length of the student's shadow x.

(ii) **[1 Mark]** If a basketball hoop pole casts a shadow of 1.2 m at that same instant, find the height of the hoop pole.

(iii) **[2 Marks]** If a 3D scale model is built with scale factor 1 : 50, find the height of the flagpole in the model and state the ratio of perimeters of any similar component.

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